

MACQUARIE University
SYDNEY·AUSTRALIA

BUSA8090 Data and Visualisation for Business

Report - Visualisations

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I. Introduction

Road safety is a significant public health issue in Victoria with substantial social and economic costs. This report analyses Victorian crash data using Tableau and R and presents five visuals to: (1) chart annual and monthly trends; (2) map regional patterns; (3) compare severity by place and time; and (4) assess links between vehicle characteristics and outcomes. Methods and data-processing steps are documented, results are clearly summarised, and ethical considerations in visualisation are addressed. The objective is to deliver actionable, evidence-based insights that inform policy and operational decisions, reducing crashes and harm.

II. Data preparation, Transformation, and Cleaning

2.1. Data sources and integration

I combined Crash, Vehicle and Person tables into a tidy model using accident and entity IDs, with separate crash-, vehicle- and person-level views to prevent double-counting.

2.2. Cleaning and transformation

Where essential fields were missing, records were excluded records missing essential fields only from the affected view; otherwise, retained values and marked them "Unknown" to preserve totals and show completeness.

2.3. Feature engineering

I binned hours into operational windows (eg., 07:00–09:00, 15:00–18:00); created a month-start date for monthly aggregation; computed vehicle age (crash year − manufacture year), capped at 50 to limit entry errors.

2.4 Geographic enrichment and aggregation

Crash points were spatially joined to Local Government Area (LGA) polygons for policy-relevant mapping. To avoid the "population trap," I merged the latest available LGA population and calculated both crash counts and a fatality rate per 1,000 residents.

III. Visualisation Analysis and Findings

3.1 Time Series Analysis of Road Crash Trends

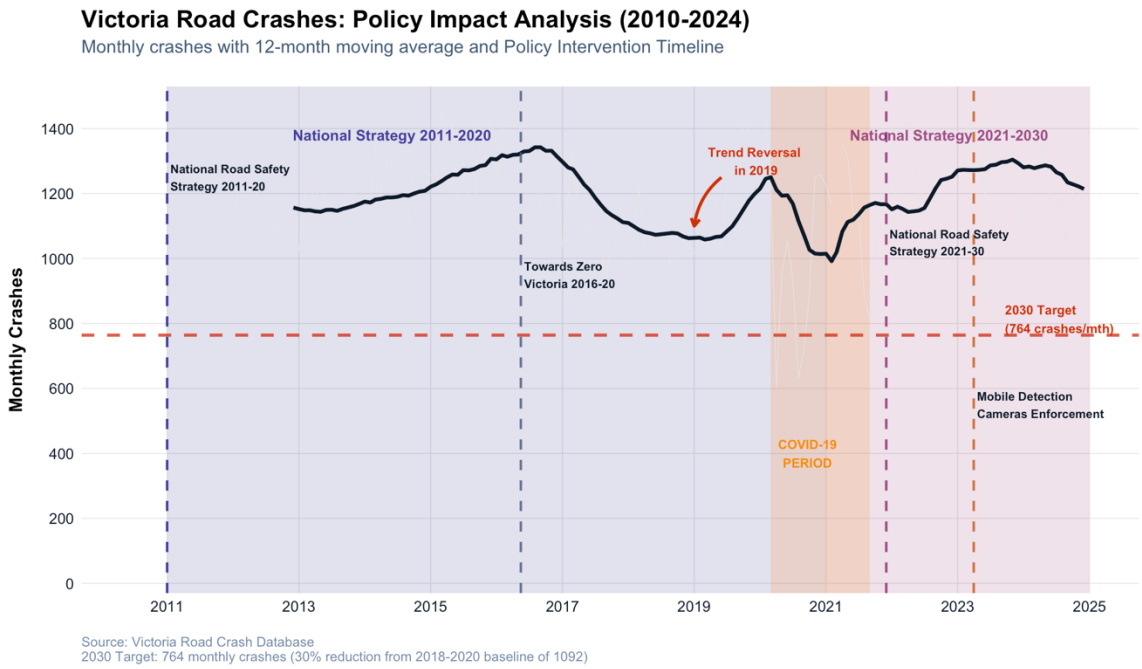


Figure 3.1: Victoria Road Crashe Trends: Policy Impact Analysis (2012-2024)

Generated by author

As shown in Figure 3.1, a dual-layer line displays raw monthly crashes to show seasonality, overlaid with a bold 12-month moving average to reveal trajectory. Smoothing volatile data to reveal signal is a cornerstone of practical time-series analysis (Few, 2009). I annotated major policy milestones, including the launch of Victoria's Towards Zero strategy (Department of Transport, 2020), and added a dashed 2030 target line from the National Road Safety Strategy (Office of Road Safety, 2021) for instant comparability.

Crashes rise to a local high around 2016–17, then decline through 2019. A sharp trough appears in 2020–21 during the COVID mobility shock, a period of unprecedented change in traffic volume and driver behaviour. From 2021, crashes rebound and stabilise near pre-pandemic levels, with slight softening in 2024. Seasonality persists winters are higher, and February is consistently lower. The current moving average is ~1,220 crashes per month, ~60% above the 2030 target of 764, a gap of roughly 450–480 crashes per month.

The mid-2010s improvement aligns with the Towards Zero strategy (Department of Transport, 2020) and gradual fleet-safety upgrades. In contrast, the pandemic drop reflects reduced exposure rather than a structural safety gain. The post-pandemic plateau suggests recent measures have stabilised outcomes but are not yet bending the curve toward the target. The pace of improvement must roughly double or triple to meet the national commitment (Office of Road Safety, 2021). Operationally, the clear seasonal signal supports concentrating prevention in high-risk months and using this series to verify whether new interventions create visible, durable inflection points.

3.2 Geographic Distribution and Risk Mapping

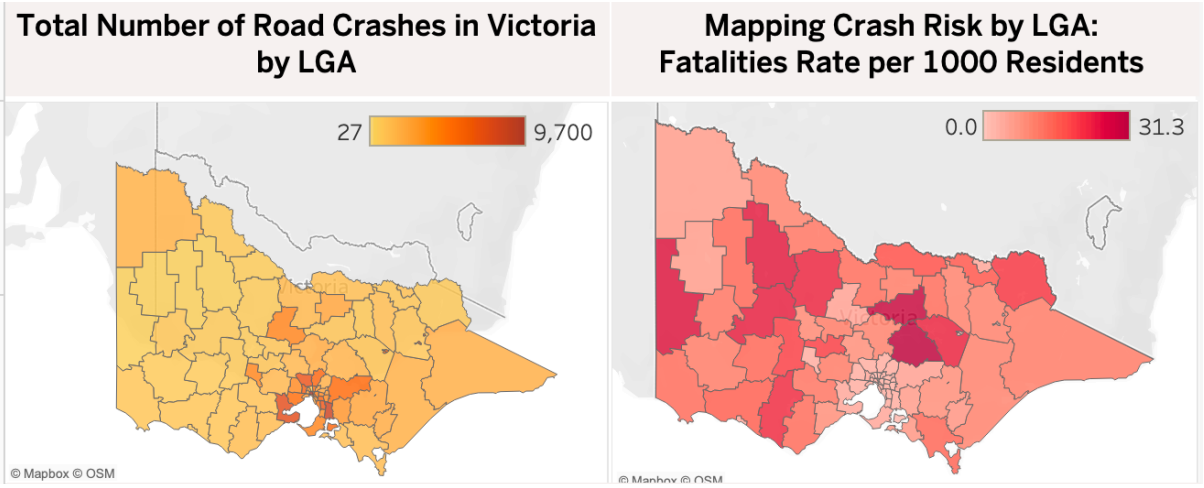


Figure 3.2: Dual View of Road Safety: Crash Volume and Fatalities per 1,000 by LGA

Generated by author

Figure 3.2 illustrates two choropleths are displayed side by side: the left shows absolute crash counts, while the right shows fatalities per 1,000 residents. A ranked bar-and-line panel underneath orders LGAs by crash count (bars, coloured Metro vs Country) and overlays the fatality rate (grey line) as displayed in Figure 3.3.

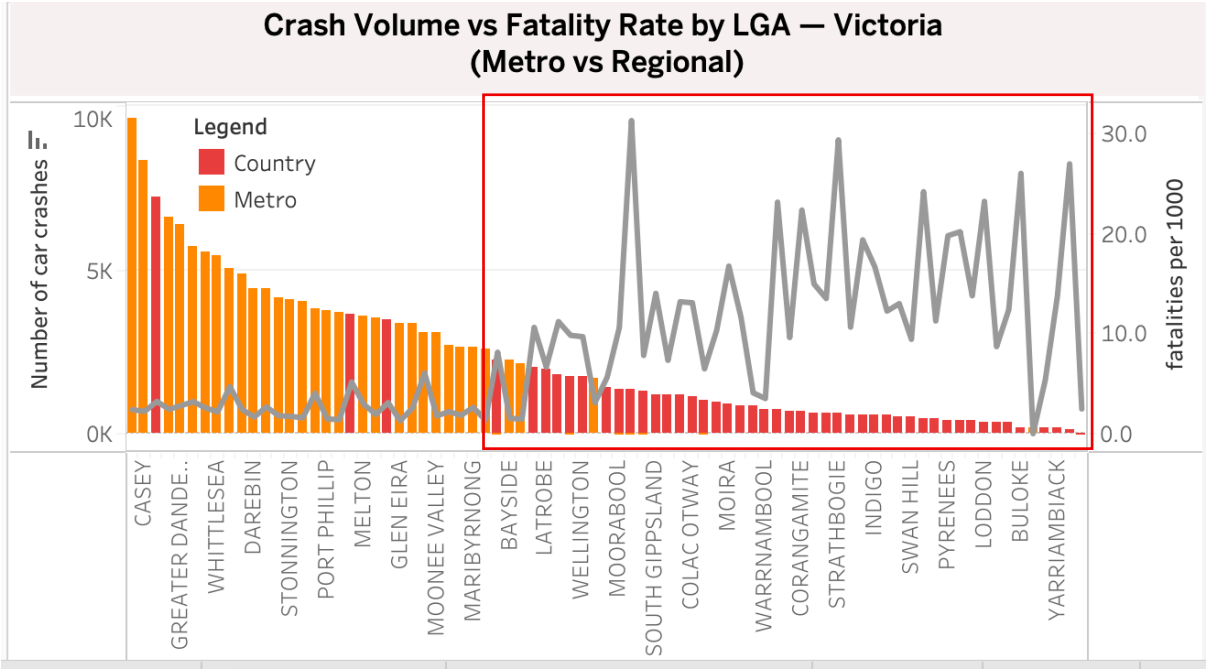


Figure 3.3: Crash Volume vs Fatality Rate by LGA — Victoria (2012–2024)

Generated by author

As demonstrated in Figure 3.3, Metropolitan LGAs (e.g., Casey, Greater Dandenong, Whittlesea) dominate absolute volumes, thousands of crashes across 2012–2024, yet their per-capita fatality rates are comparatively low. Several regional LGAs (e.g., Buloke, Yarriambiack, Loddon) show the opposite pattern: small counts but high fatality rates (spiking to the right of the ranking chart). This "inversion" makes clear that high volume does not necessarily equal high per-capita risk.

Mechanisms differ by geography. In metro areas, crashes are a frequency/exposure problem (dense traffic, intersections, distraction) with lower average speeds limiting severity. In regional areas, crashes are a severity/survivability problem (higher speeds, unforgiving roadsides, and more extended emergency response times). Policy should therefore diverge:

- Metro: Reduce conflict/volume by speeding compliance, improving intersection treatments, and managing congestion and distraction.
- Regional: reduce severity: median barriers and shoulder sealing, speed management on high-speed links, improved trauma response coverage, and accelerated uptake of safer vehicles.

A count-only map can stigmatise large LGAs; a rate-only map hides service burden. Showing both maps, then reinforcing the view with the ranked bar-plus-rate line, provides a balanced perspective that directly supports differentiated investment.

3.3 Crash Severity by Time and Demographics

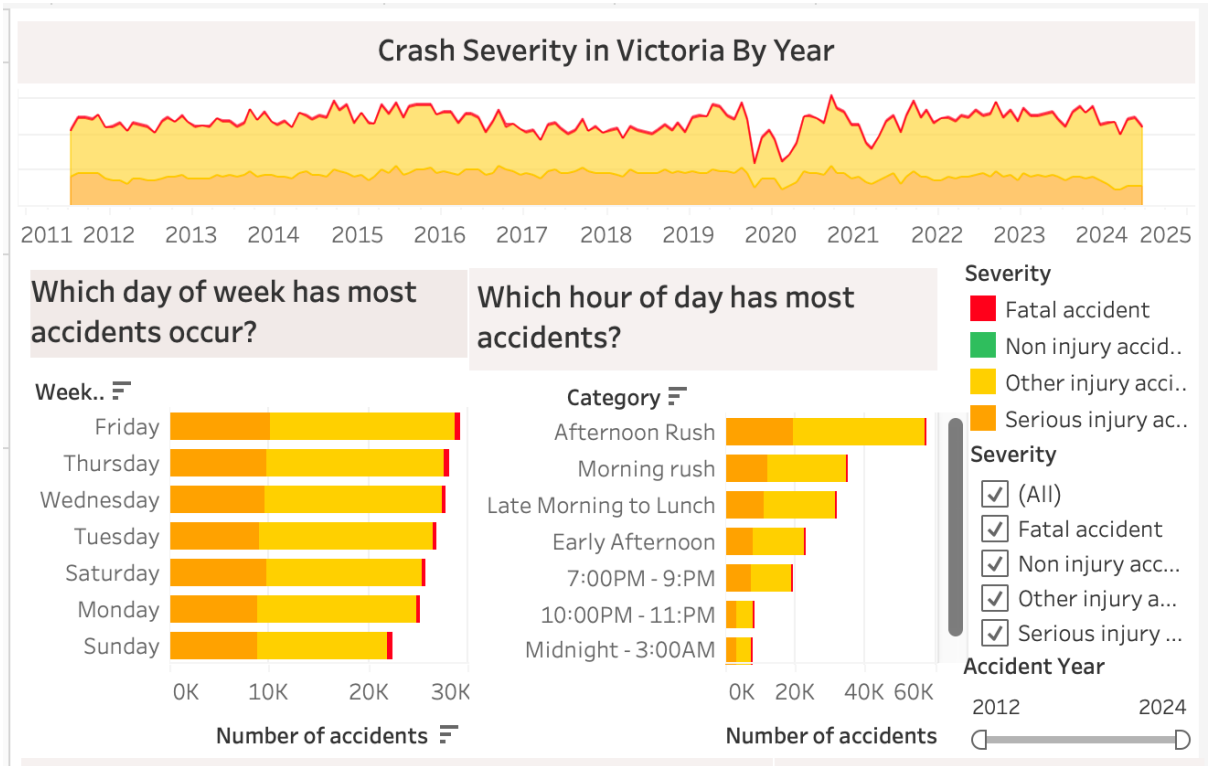


Figure 3.4: Crash Severity in Victoria by Year, Weekday, and Hour (2012–2024)

Generated by author

This dashboard brings together the "when" and the "who". As shown in Figure 3.4, a stacked area traces total crashes alongside severity composition over time, while ranked bars display peak periods by weekday and hour. As displayed in Figure 3.5, a diverging population-pyramid compares male and female casualties across age bands, with a small panel summarising the overall gender split.

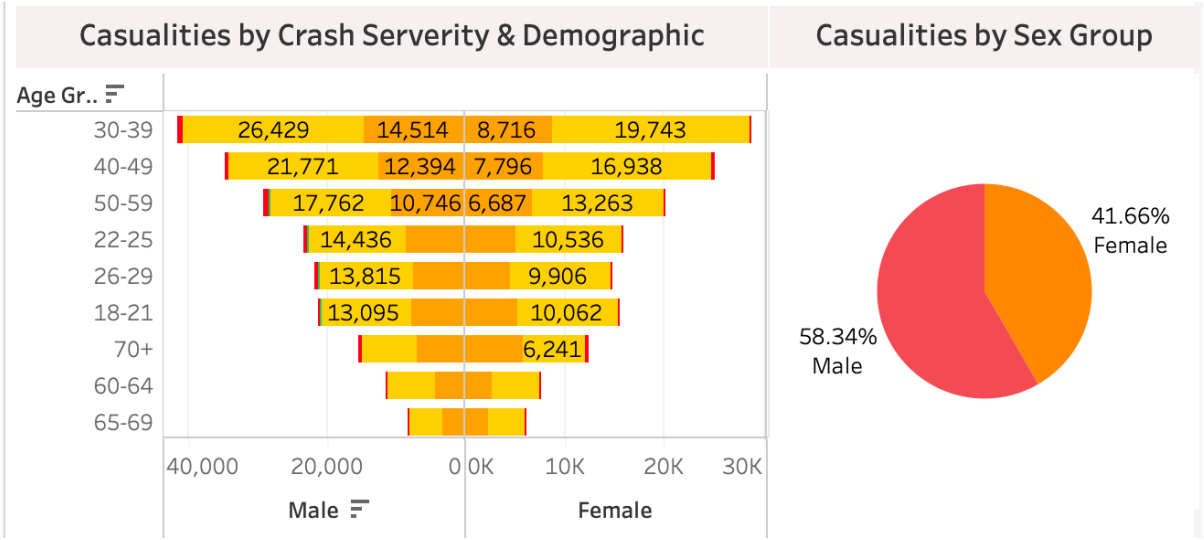


Figure 3.5: Crash Severity in Victoria by Demographic (2012–2024)

Generated by author

As shown in Figure 3.4, Fridays record the highest crash counts and Sundays the lowest. The afternoon rush (3–6 pm) dominates the daily profile, roughly two to three times the morning peak, and, when combined with Fridays, marks the highest-risk window. Demographically, as shown in Figure 3.5, males account for approximately 58.3% of casualties and are overrepresented in every age group; the 30–39 cohort bears the most considerable absolute burden, consistent with the peak of commuting and family travel. Despite large swings in total volume, especially the COVID-19 trough in 2020, the proportional mix of severity remains stable: serious injury and other-injury incidents dominate, non-injury is relatively minor, and fatal crashes form a relatively thin band.

The Friday-afternoon peak likely reflects heavier exposure, end-of-week fatigue, and distraction (see Figure 3.4). Male over-representation aligns with higher driving exposure and riskier behaviours (see Figure 3.5). Because the severity mix remains stable even when volumes decline, policies that only reduce exposure may not alter outcomes. Targeted severity-reduction measures, such as speed management, safer roadways and vehicles, and faster trauma response, should be accompanied by peak-time enforcement and effective communication.

3.4 Crash Severity by Vehicle-Related Factors

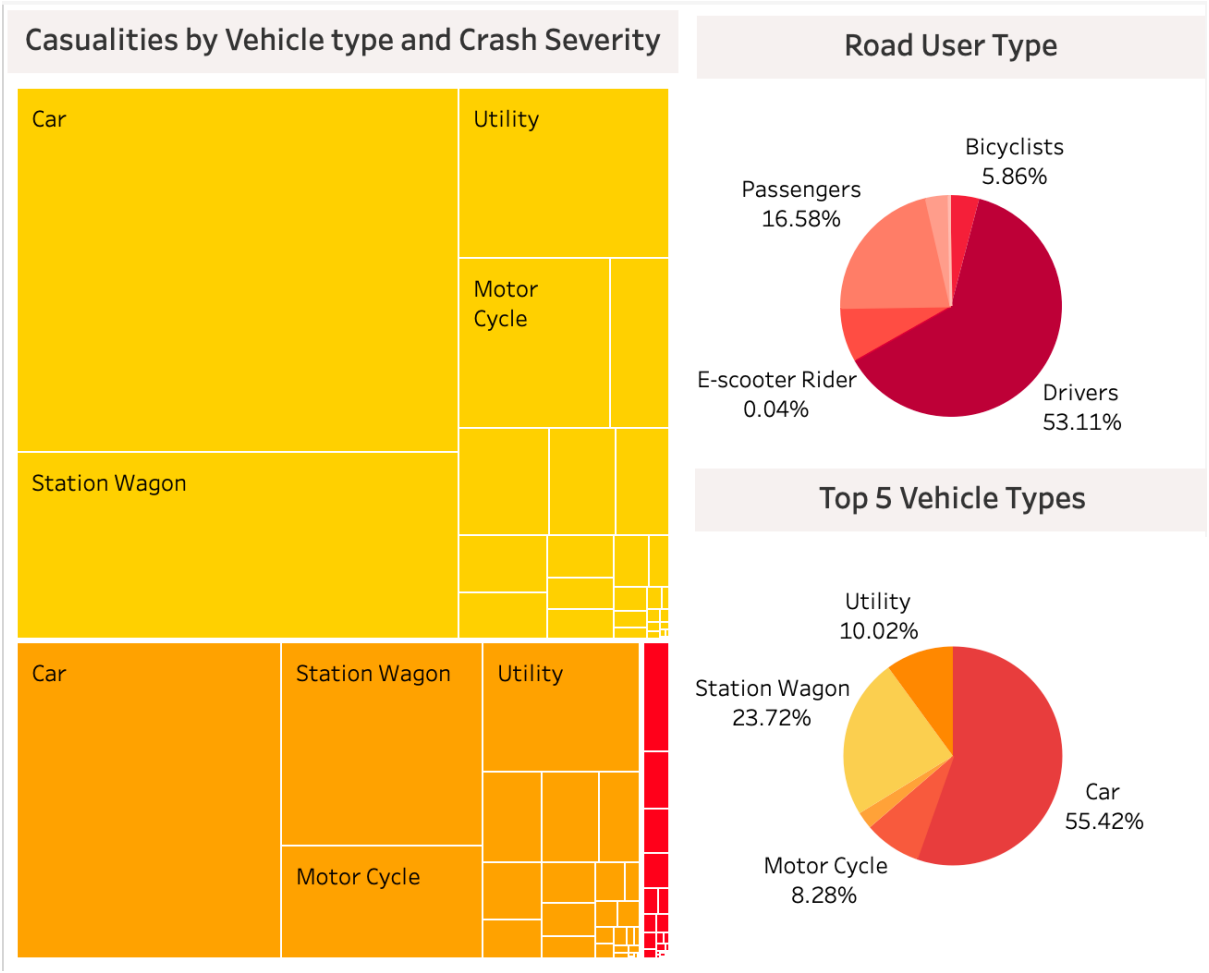


Figure 3.6: Vehicle type & road-user role vs crash severity

Generated by author

As shown in Figure 3.6, the treemap plots casualties by vehicle type (area) and by severity (colour). Two small pie charts summarise road-user roles and the top vehicle types. As shown in Figure 3.7, a horizontal bar chart illustrates the relationship between crash severity and the average age of the vehicles involved in the crashes.

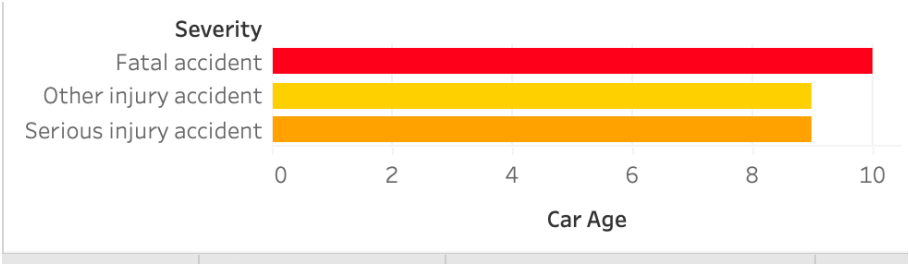


Figure 3.7: Vehicle Age and Crash Severity in Victoria

Generated by author

As shown in Figure 3.6, cars (55.4%) and station wagons (23.7%) account for the majority of casualties; however, their blocks are mostly yellow, indicating less severe outcomes. Motorcycles (8.3%) and utilities (10.0%) appear less frequently, but they are associated with many serious and fatal outcomes. For road-user roles, drivers account for the largest share (53.1%), followed by passengers (16.6%) and bicyclists (5.9%). As shown in Figure 3.7, fatal crashes typically involve older vehicles, on average, around ten years old, while serious and other-injury crashes involve slightly newer vehicles.

These patterns point to simple actions. Motorcycles lack a protective shell, and some utilities have a higher risk of rollover, so severity concentrates there even when counts do not. Newer cars come equipped with more safety features, so renewing the oldest part of the fleet should reduce deaths. Pair high-frequency measures for cars (speed and distraction management) with high-severity measures for motorcycles and utilities (protective gear, forgiving roadsides). Then, prioritise policies that accelerate the replacement of older vehicles.

IV. Ethical Considerations in Data Visualisation

4.1 Transparency and accuracy

Charts use truthful encodings and precise scales. Bar charts start at zero, and dual-axis views are avoided unless fully labelled. Where smoothing is applied (the 12-month moving average), it is disclosed on the graphic and in the caption. Policy makers are annotated to prevent readers from confusing correlation with causation. As shown in Figures 3.2 and 3.3, counts and rates are displayed side by side to minimise the risk of misreading due to volume alone.

4.2 Privacy and harm minimisation

All outputs use aggregated data. Maps are shown at the LGA level, not exact coordinates, and tiny cells or rare categories are suppressed or rolled into "Other/Unknown." No personally identifiable information appears anywhere in the report. This protects individuals and communities while preserving analytical value.

4.3 Balanced, evidence-based representation (with accessibility)

Visuals are designed to provide context rather than sensationalise. As displayed in Figures 3.6 and 3.7, vehicle analysis reveals both frequency and severity, allowing low-volume, high-severity groups (e.g., motorcycles) to be visible without blaming communities or users. Colour palettes are consistent across severity categories and chosen for legibility, including common colour-vision deficiencies. Titles, axes, and legends are written in plain language.

Derived fields, such as moving average, fatalities per 1,000 residents, and severity groupings, are defined once and reused consistently. Filters are displayed on each dashboard, and transformation choices (for example, hour bins) are clearly stated so that others can reproduce, audit, or extend the analysis. These practices keep the work truthful, privacy-preserving, and decision-ready.

V. Conclusion

This report synthesises five visuals on road safety in Victoria. Crash counts fell in the late 2010s, then collapsed during the pandemic due to lower mobility, before rebounding from 2021 and remaining above the 2030 target.

Geography separates exposure from risk: metropolitan LGAs record most crashes, while regional LGAs have higher fatality rates per person. Time patterns matter: Friday afternoons are the busiest, and late nights carry a larger share of severe outcomes despite fewer crashes. Males are overrepresented; adults aged 30 to 39 carry the heaviest burden. Motorcycles, utility vehicles, and older vehicles are more frequently involved in serious or fatal crashes.

Priorities are clear: in metro areas, reduce intersection conflicts and improve speed compliance; in regional areas, lower speeds, add forgiving road shoulders and barriers, and strengthen trauma response; allocate targeted resources to Friday afternoons and late nights; accelerate fleet renewal. Reports include rates, disclose uncertainty, and design visuals that avoid misreading.

VI. References:

Few, S. (2009). Now You See It: Simple Visualization Techniques for Quantitative Analysis. Analytics Press. Tufte, E. R. (2001). The Visual Display of Quantitative Information (2nd ed.). Graphics Press.

Department of Transport (Victoria). (2020). *Victorian road safety strategy 2021–2030. State of Victoria (Department of Transport)*.https://www.tac.vic.gov.au/road-safety/victorian-road-safety-strategy/victorian-road-safety-strategy-2021-2030/RoadSafetyStrategy_DEC2020.pdf

Office of Road Safety. (2021). *National Road Safety Strategy 2021–2030*. Australian Government Department of Infrastructure, Transport, Regional Development and Communications. https://www.roadsafety.gov.au/nrss